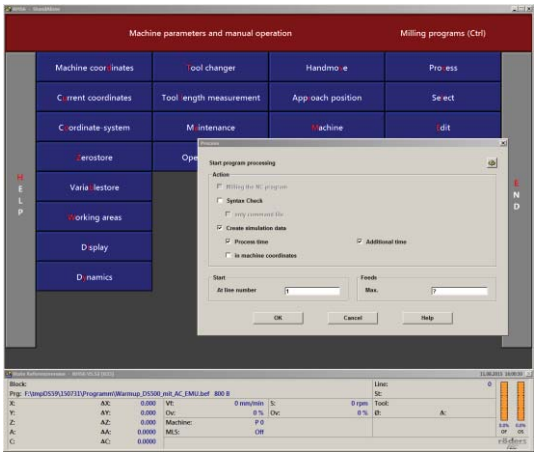
RMS6 - ERP
(ERP-State)

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2 History

■ 1.01	Original version, based on TC.0919	D. Wagner
■ 1.02	Layout	D. Wagner
■ 1.03	Versionstate	S. Friedemann
■ 1.04	Expansion <state> and outputs <doorstate>, <autodoorstate> and <lastjobend>	S. Friedemann
■ 1.05	English translation	M.Reinecke
■ 1.06	Expansion BDE output	S. Friedemann
■ 1.07	Divided into interface and setup documentation	S. Dailidow
■ 1.08	Multiple spindles and location	H. Sieck
■ 1.09	Layout	S. Friedemann
■ 1.10	<tooldoorstate>	D. Wagner
■ 1.11	Change variable names for waiting times	S. Friedemann
■ 1.12	Expansion BDE output	S. Friedemann

3 Version state

- Generating ERP state

  Since RMS6 version: 3.50 (196)

  3.60 (207)

- Creation of additional outputs concerning state of doors (manuel and automatic) and time of last processing

  Since RMS6 version: 4.61 (246) and 4.62 (125)

- Differentiation of machinestate „IDLE“ regarding ‘machine OUT’ or ‘machine ECO Mode[1]‘

  Since RMS6 version: 4.61 (246) and 4.62 (125)

- Option: additional output of operation datat (Machine.bde, c_MDC, s4ERPStateOption=&2))

  Since RMS6 version: 5.51 (236)

- Option: Multiple spindles and location

  Since RMS6 version: 5.54 (124)

- Output nominal torque of the spindle in percent

  Since RMS6 version: 6.22 (345) und 6.23 (293)

4 File output – ERP State

If the RMS6-control is active, machine state is outputted. In order to do so an XML-file is updated cyclical.

4.1 Time format

- Time format has to be *YYYY-MM-DD HH:MM:SS*[2].

  Example: 2010-02-23 09:27:55

- Time counter in seconds

4.2 Data structure

The parts of *xml*-file beeing used for the data output and their meaning is going to be explained in the following.

[1] Additional configuration of the RMS6 is necessary
[2] time disclosures: Y = year; M = month; D = day; H = hour; M = minute; S = second

4.2.1 XML-document

- All names of elements are in lowercase and can contain a text content, for empty elements you can use the abbreviated notation.

  The elements are free of attributs.

  The root-element is called <*erp*> and is above an ERP-output block.
- Encoding is done according to ISO-8859-1

4.2.2 XML-contents

- The table includes names of elements,their usage, and also the used output formats.

Element	Usage	Data type
<erp>	Root-elelement, is ahead of an ERP-output block	-
<version>	Version of output format	Float

<data>	xml-block for data	
<time>	Timestamp	Time format: YYYY-MM-DD HH:MM:SS
<name>	Machine name	Text
<location>[3]	Machine location	Text
<cluster>[4]	Machine cluster	Text
<ver>	Version of RMS6-controlsoftware	Text
<mode>	Mode **MANUAL, AUTOMATIC**	Text
<state>[5]	State of machine **IDLE, MOFF, ECO, ENHECO RUNNING, ERROR, NOT_CONNECTED**	Text

[3] since RMS6 5.54 (124)
[4] since RMS6 5.54 (124)
[5] The states „OFF“ and „ECO“ are optional (additional RMS6 configuration is necessary, since RMS6 4.61 (246) or 4.62 (125))

<doorstate>[6]	State of operator machine door	Text
	CLOSED, OPEN	
<autodoorstate>[6]	State of automatic door	Text
	CLOSED, OPEN	
<tooldoorstate>[7]	State of toolchanger automatic door	Text
	CLOSED, OPEN	

<feed>	Feed (mm/min)	Float
<ov_feed>	Override feed (in percent)	Float
<tool>	Locationnumber of actual tool	Integer
<tooltype>	Spindel tooltype	Text
<spinspeed>	Spindle speed (rpm) of current spindle	Float
<spinforce>	Nominal torque of the spindle (in percent)[8]	Float
<ov_spinspeed>	Override spindle speed (in percent, current spindle)	Float

<job>	xml-block for actual job(only during a job is running)	
<name>	Current job	Text
<ncpgm>	Path of chosen NC-file	Text
<start>	Time of jobstart	Time format: YYYY-MM-DD HH:MM:SS
<lastjobend>[9]	Time of completing last job	Time format: YYYY-MM-DD HH:MM:SS
<subpgm>	Path of the actual subprogram	Text
<substart>	Time of beginning subprogram	Time format: YYYY-MM-DD HH:MM:SS
<percent>	Progress (in percent)	Float
<line>	Current program line	Integer

[6] The element is available since RMS6 4.61 (246) or 4.62 (125)
[7] The element is available since RMS6 5.74 (070)
[8] The element is available since RMS6 6.22 (345) or 6.23 (293)
[9] Element is available since RMS6 4.61 (246) or 4.62 (125)

<chuck>	Current Chuck	Integer
<pallet>	Current pallet	Integer
</job>	End of xml-block *job*	

<user>	xml-block, user-specific data, see chapter 5.2.1	
</user>	End xml-block *user*	

<bde>	xml-Block for operating data	
<bde_start>	Start of operating data collecting	Time format: JJJJ-MM-TT HH:MM:SS
<op_duration>	Operating time	Time in seconds
<op_percent>	Operating utilization in percent	Text
<nc_duration>	Milling time	Time in seconds
<nc_percent>	Milling utilization in percent	Text
<set_duration>	Rüstzeit	Time in seconds
<set_percent>	Setup time utilization in percent	Text
<spin_start>	Start of spindle data collecting of first spindle	Time format: JJJJ-MM-TT HH:MM:SS
<spin_duration>	Spindle on time of first spindle	Time in seconds
<spin2_start>[10]	Start of spindle data collecting of second spindle	Time format: JJJJ-MM-TT HH:MM:SS
<spin2_duration>[11]	Spindle on time of second spindle	Time in seconds
</bde>	End xml-Block *bde*	

</data>	End xml-block *data*	
</erp>	End xml-block *erp*	

[10] since RMS6 5.54 (124)
[11] since RMS6 5.54 (124)

4.2.1 User-specific output fields

- By NC-command *ERPSTATE* with the options *–N*, *–NUM* / *-T* you can get an userspecific output while the program is in process.

  Opening processing dialog will delete the whole content of user specific output. By specifying the same name with an empty text in output the entry for this name will be erased.

Element	Usage	Data type
<user>	xml-block for user-specific data	
<?>		
</user>	End xml-block *user*	

Syntax

ERPSTATE *-N*= [*-T*= or *-Num*=]

Options:

-N=	Sets name for output.
-T=	Sets text for output.
-Num=	Sets value for output.

4.3 File access

The output file will be updated automaticlly if data has been changed. Updating is done (c_MDC, f4*ERPRefreshFile*[12]) in seconds.

Requests to the file must be selected, the file has to be released in RMS6-settings (0.5*f4*ERPRefreshFile*), otherwise an error message will appear.

If the filepath is not valid an error message will appear.

[12] Old: s2ERPRefreshFile

5 Examples

- Machine in idle state

Example: ERP-State (state.xml)

```
<?xml version="1.0" encoding="iso-8859-1"?>

<erp>
   <version>1.0</version>
   <data>
      <time>2008-03-05 10:44:53</time>
      <name>RXP500_Test</name>
      <ver>3.41 (212)</ver>
      <mode>MANUAL</mode>
      <state>IDLE</state>
      <feed>0</feed>
      <ov_feed>100</ov_feed>
      <tool />
      <tooltype />
      <spinspeed>0</spinspeed>
      <ov_spinspeed>100</ov_spinspeed>
      <job>
         <ncpgm>f:\public\pro\testup\hp.bef</ncpgm>
      </job>
      <user />
   </data>
</erp>
```

- Machine with running job

Example: ERP-State (state.xml)

```
<?xml version="1.0" encoding="iso-8859-1"?>

<erp>
   <version>1.0</version>
   <data>
      <time>2008-03-05 10:44:53</time>
      <name>RXP500_Test</name>
      <ver>3.41 (212)</ver>
      <mode>MANUAL</mode>
      <state>RUNNING</state>
      <feed>2500</feed>
      <ov_feed>100</ov_feed>
      <tool>3</tool>
      <tooltype>Typ1</tooltype>
      <spinspeed>3000</spinspeed>
      <ov_spinspeed>100</ov_spinspeed>
      <job>
         <name />
         <ncpgm>f:\public\pro\testup\hp.bef</ncpgm>
         <start>2008-02-06 10:08:43</start>
         <subpgm>f:\public\pro\testup\up.bef</subpgm>
         <substart>2008-02-06 10:09:14</substart>
         <percent>4</percent>
         <line>10</line>
         <chuck />
         <pallet />
      </job>
      <user />
   </data>
</erp>
```

■ Machine with running job and user output

Example: ERP-State (state.xml)

```
<?xml version="1.0" encoding="iso-8859-1"?>

<erp>
   <version>1.0</version>
   <data>
      <time>2009-01-29 07:53:07</time>
      <name>RXP500_Test</name>
      <ver>3.60 (260)</ver>
      <mode>MANUAL</mode>
      <state>RUNNING</state>
      <feed>2500</feed>
      <ov_feed>100</ov_feed>
      <tool>1</tool>
      <tooltype>Typ1</tooltype>
      <spinspeed>5000</spinspeed>
      <ov_spinspeed>100</ov_spinspeed>
      <job>
         <name />
         <ncpgm>D:\test1.bef</ncpgm>
         <start>2009-01-29 07:52:28</start>
         <percent>9%</percent>
         <line>8</line>
         <chuck />
         <pallet />
      </job>
      <user>
         <text1>2233475748.0000</text1>
         <text2>Order 4711</text2>
      </user>
   </data>
</erp>
```

■ Syntax for an example of user output

Example: Text excerpt NC-program

```
ERPSTATE -N="text1"  -NUM=2233475748
…
ERPSTATE -N="text2"  -T="Order 4711"
...
ERPSTATE -N=text1  -T=""                    // delete entry text1
```